

CPSC 340: Machine Learning and Data Mining

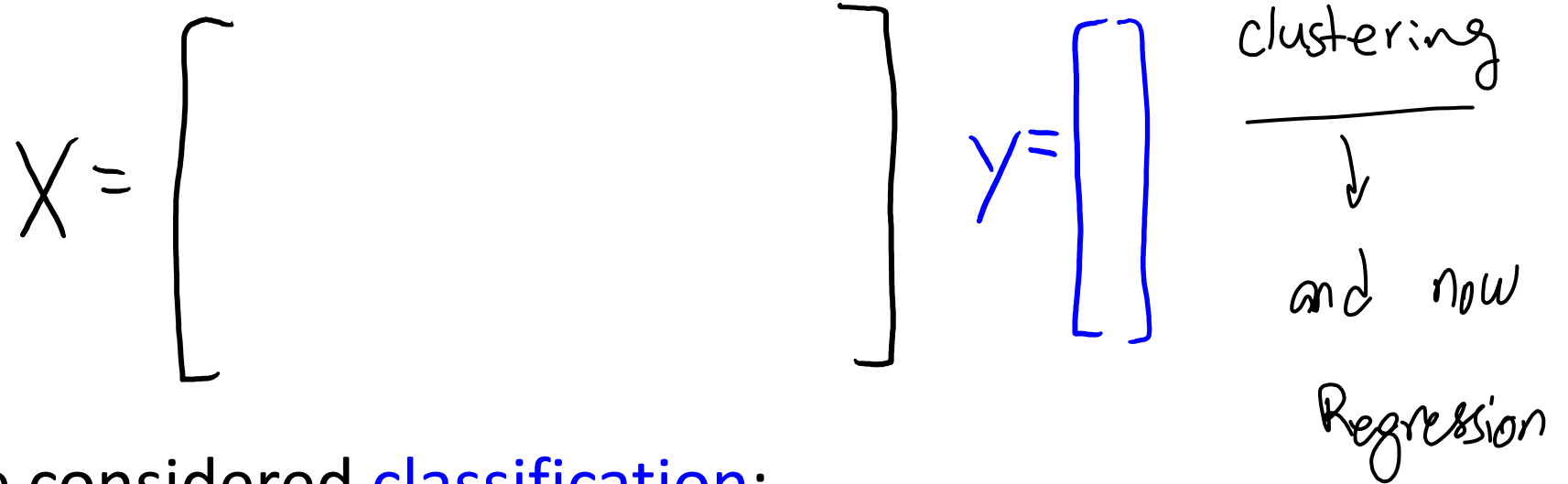
Linear Regression

Admin

- **Assignment 2:**
 - Due on Monday.
 - 1 late class to hand in Wednesday, 2 for next Friday.
- **Midterm:**
 - Room: IRC 2
 - Tuesday, 2025-10-21 at 17:00-19:00
 - Conflicts: email cpssc340-admin@cs.ubc.ca (the weight will be put on the final).
- We're going to start using **calculus** and **linear algebra** a lot.
 - You should **start reviewing these ASAP** if you are rusty.
 - A review of relevant calculus concepts is [here](#).
 - A review of relevant linear algebra concepts is [here](#).

Supervised Learning Round 2: Regression

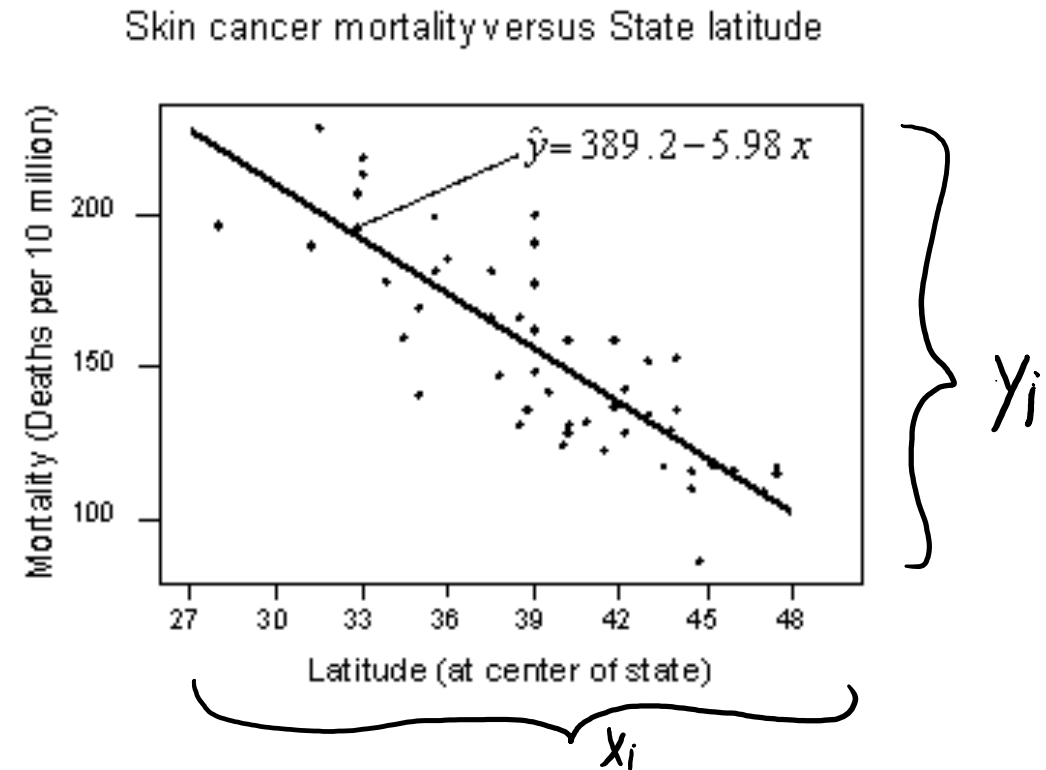
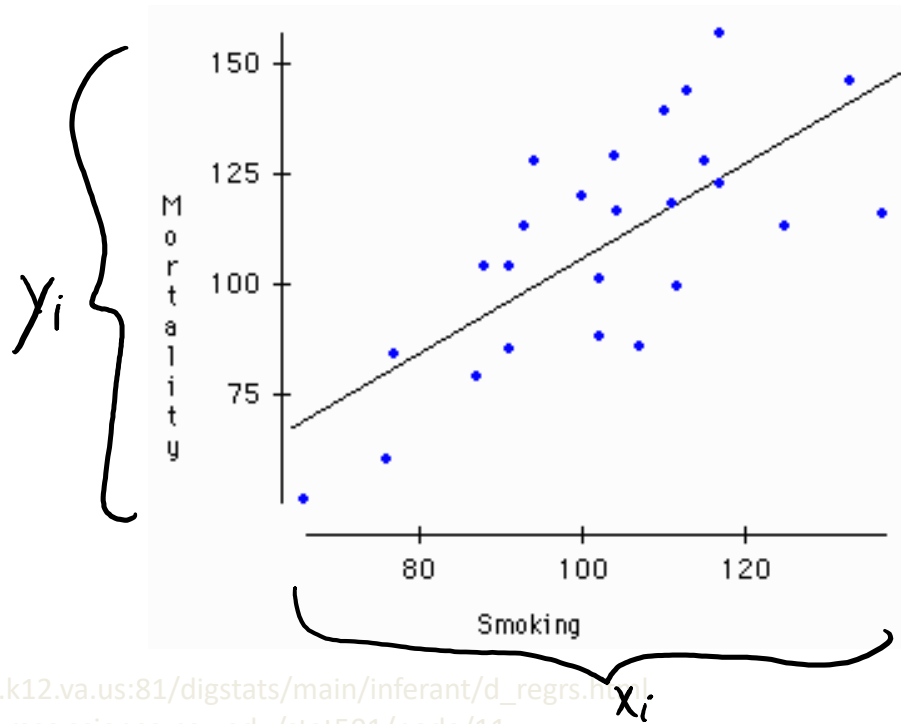
- We're going to revisit supervised learning:



- Previously, we considered classification:
 - We assumed y_i was categorical: $y_i = \text{'spam'}$ or $y_i = \text{'not spam'}$.
- Now we are going to consider regression:
 - We allow y_i to be numerical: $y_i = 10.34\text{cm}$.

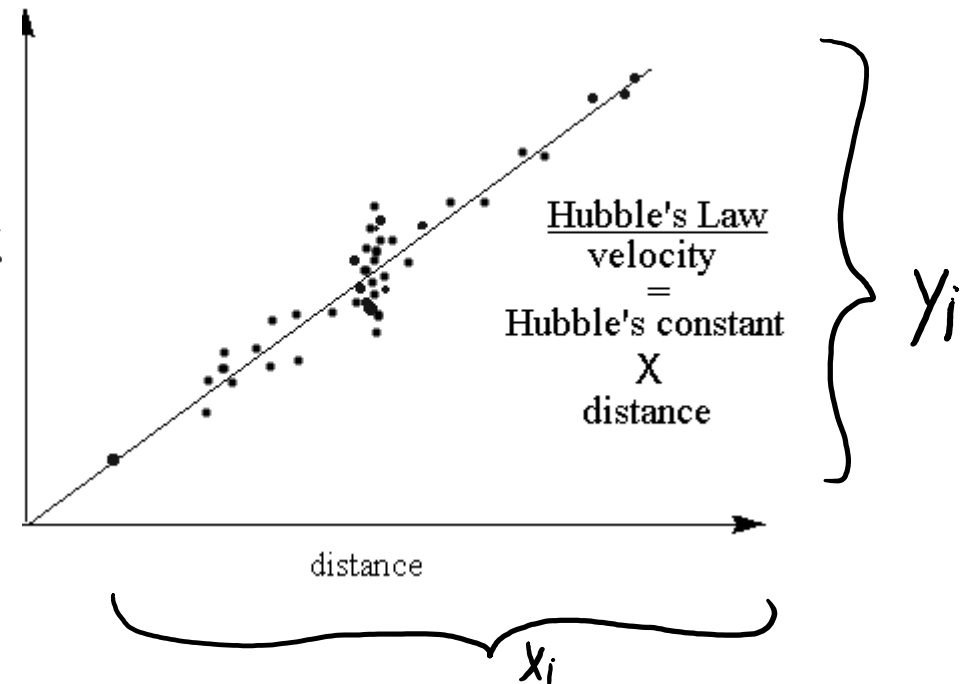
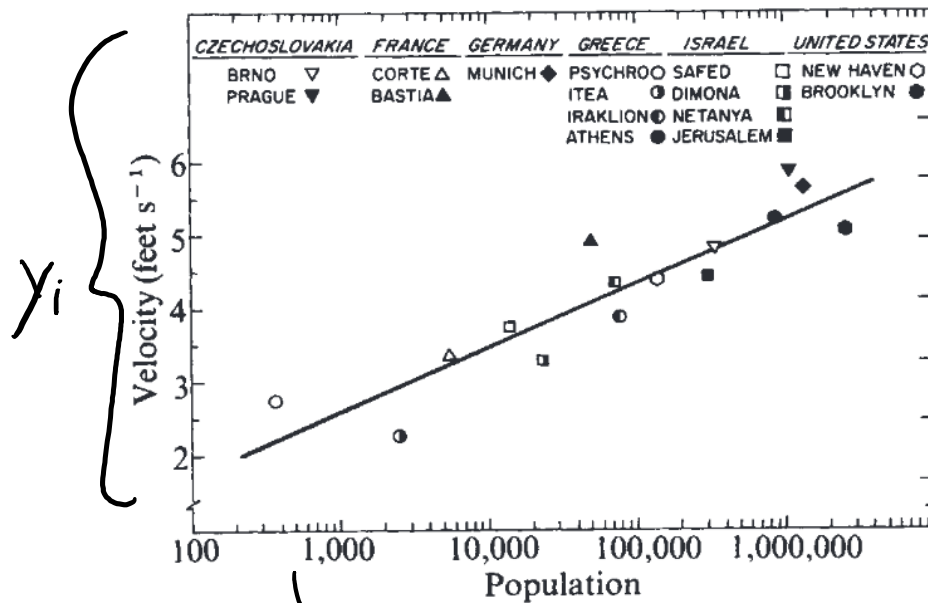
Example: Dependent vs. Explanatory Variables

- We want to **predict a numerical value** given features:
 - Does number of lung cancer deaths change with number of cigarettes?
 - Does number of skin cancer deaths change with latitude?



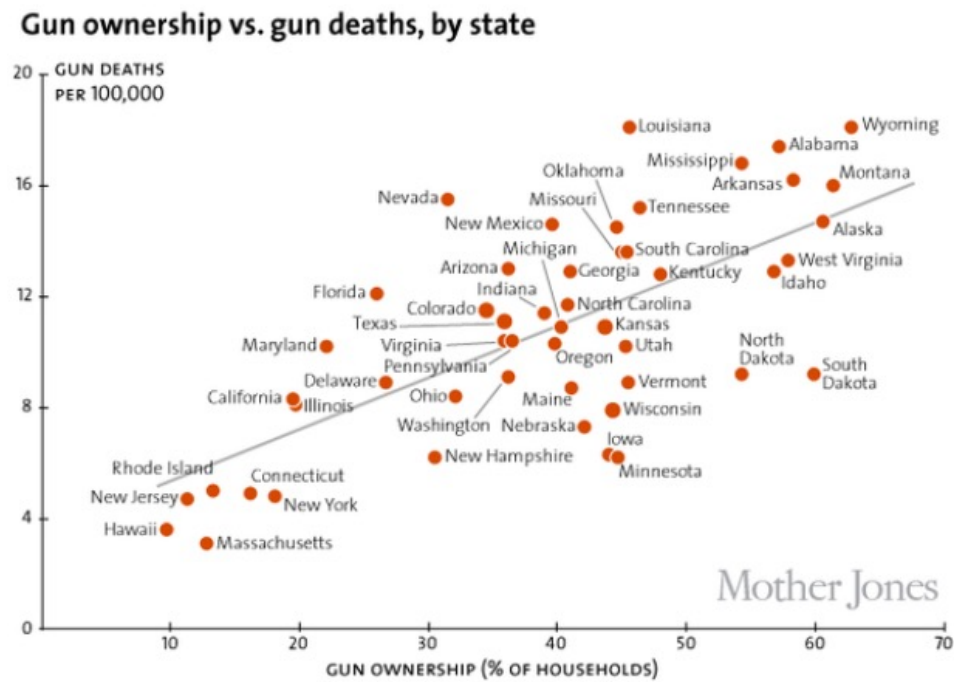
Example: Dependent vs. Explanatory Variables

- We want to **predict a numerical value** given features:
 - Do people in big cities walk faster?
 - Is the universe expanding or shrinking or staying the same size?



Example: Dependent vs. Explanatory Variables

- We want to **predict a numerical value** given features:
 - Does number of gun deaths change with gun ownership?
 - Does number violent crimes change with violent video games?

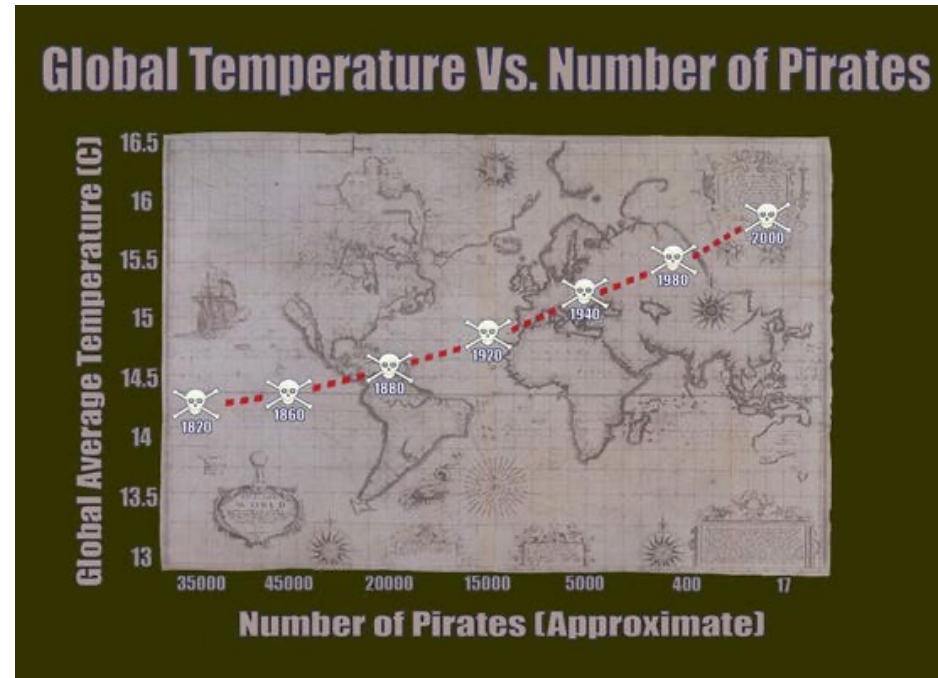


Spurious Associations

- Does number of pirates help predict global temperature?

- **Spurious association.**

- Variables that seem related, but this is a coincidence.

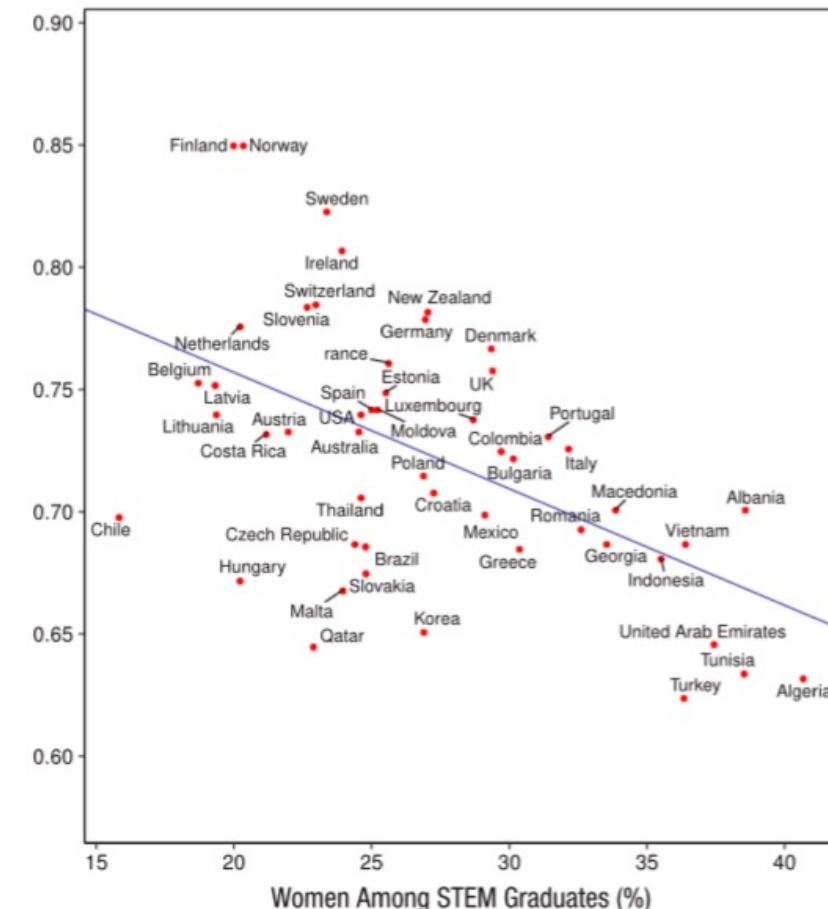


- If comparing enough variables, can get strong relationships by chance.
- Fun with spurious correlations [here](https://www.forbes.com/sites/erikaandersen/2012/03/23/true-fact-the-lack-of-pirates-is-causing-global-warming).

یہ خدائے باری ہے

Example: Dependent vs. Explanatory Variables

- We want to **predict a numerical value** given features:
 - Does higher gender equality index lead to more women STEM grads?
- Note that we are doing **supervised** learning:
 - Trying to predict value of 1 variable (the ' y_i ' values). (instead of measuring correlation between 2).
- Supervised learning **does not give causality**:
 - OK: “Higher index **is correlated** with lower grad %”.
 - OK: “Higher index **helps predict** lower grad %”.
 - BAD: “Higher index **leads to** lower grads %”.
 - People/media get these confused all the time, be careful!
 - There **are lots of potential reasons for this correlation**.



Handling Numerical Labels

- One way to handle numerical y_i : **discretize**.
 - E.g., for ‘age’ could we use {‘age ≤ 20 ’, ‘ $20 < \text{age} \leq 30$ ’, ‘age > 30 ’}.
 - Now we can apply methods for classification to do regression.
 - But **coarse discretization loses resolution and order**.
 - And **fine discretization requires lots of data** (“coupon collecting”).
- There exist regression versions of classification methods:
 - Regression trees, neighbour-based methods, and so on.
- Today: one of oldest, but still most popular/important methods:
 - **Linear regression based on squared error**.
 - Interpretable and the **building block** for more-complex methods.

Least Squares with one Feature

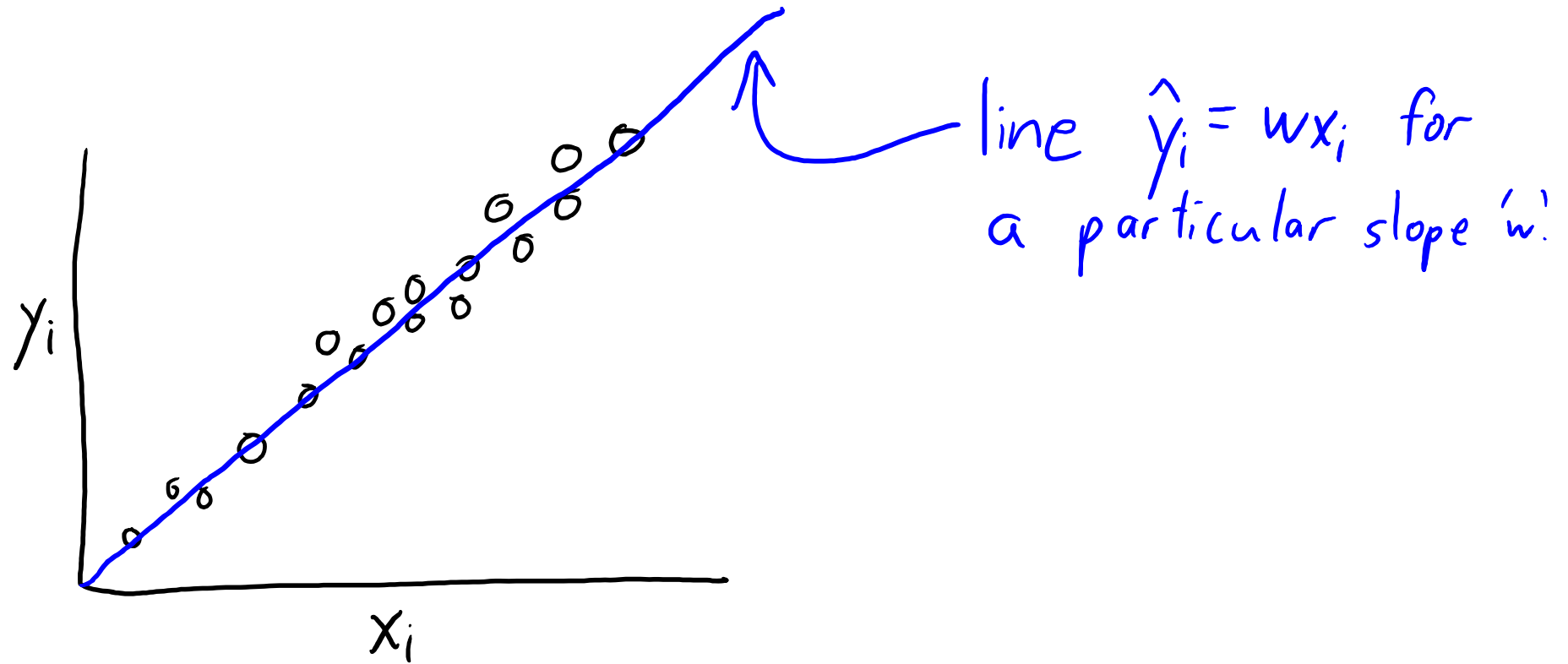
Linear Regression in 1 Dimension

- Assume we only have 1 feature ($d = 1$):
 - E.g., x_i is number of cigarettes and y_i is number of lung cancer deaths.
- **Linear regression** makes predictions $\hat{y}_i$ using a **linear function** of x_i :

$$\hat{y}_i = w x_i$$

- The parameter 'w' is the **weight** or **regression coefficient** of x_i .
 - We are temporarily ignoring the y-intercept.
- As x_i changes, slope 'w' affects the rate that $\hat{y}_i$ increases/decreases:
 - Positive 'w': $\hat{y}_i$ increase as x_i increases.
 - Negative 'w': $\hat{y}_i$ decreases as x_i increases.

Linear Regression in 1 Dimension



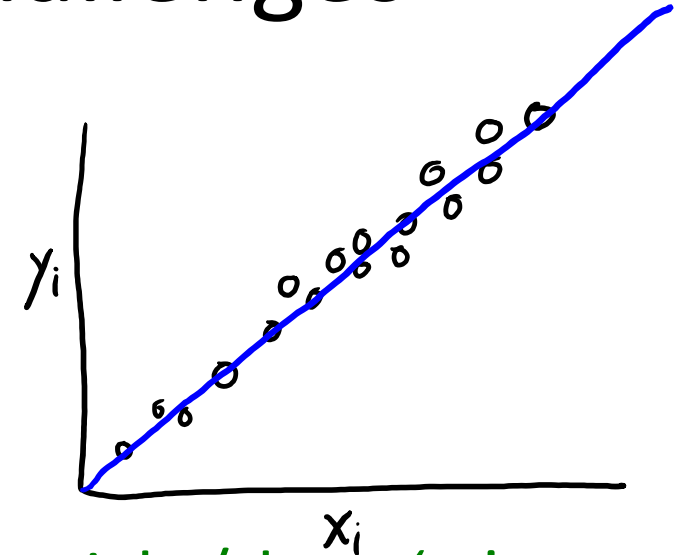
Aside: terminology woes

- Different fields use different terminology and symbols.
 - Data points ' i ' = **objects** = **examples** = rows = observations.
 - **Inputs** x_i = predictors = **features** = explanatory variables = regressors = independent variables = covariates = columns.
 - **Outputs** y_i = outcomes = targets = response variables = dependent variables (also called a "label" if it's categorical).
 - Regression coefficients ' w ' = **weights** = parameters = betas.
- With linear regression, the symbols are inconsistent too:
 - In ML, the data is X and y , and the weights are w .
 - In statistics, the data is X and y , and the weights are β .
 - In optimization, the data is A and b , and the weights are x .

Linear Regression Training Challenges

- Linear regression makes **predictions** by using:

$$\hat{y}_i = w \tilde{x}_i$$



- To train a linear regression model, we **need to find weight/slope 'w'**.
- Challenges** in finding 'w' compared to fitting a decision stump:
 - Cannot **enumerate all possible values** of 'w' (could be any real number).
 - Instead, we will use calculus to find the best 'w'.
 - It is **unlikely that a line will go exactly through many data points**.
 - Due to noise, relationship not being quite linear, or just floating-point issues.
 - So it **does not make sense to find the 'w' minimizing how many times $\hat{y}_i \neq y_i$** .

Residuals and Sum of Squared Residuals

- The **residual** is the difference between our prediction and true value:

$$r_i = \hat{y}_i - y_i$$

- This can be positive or negative.
 - If this is **close to zero**, then our prediction is **close** to the true value.
- We typically **look for a 'w' that makes residuals close to zero**.
 - For example, many models minimize the **sum of the squared residuals**:
$$(\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 + \dots + (\hat{y}_n - y_n)^2$$
 - The smaller we make this, the smaller the distance between our predictions and targets.
 - Plugging in $\hat{y}_i = wx_i$ for the case of **linear** regression, we get:
$$(wx_1 - y_1)^2 + (wx_2 - y_2)^2 + \dots + (wx_n - y_n)^2$$

- The **linear least squares** model minimizes this function to choose the slope 'w'.

Linear Least Squares Objective Function

- Linear least squares sets 'w' is to minimize **sum of squared residuals**:

$$f(w) = \sum_{i=1}^n (w x_i - y_i)^2$$

True value of y_i

Our prediction $\hat{y}_i$

Sum up the squared differences over all training examples.

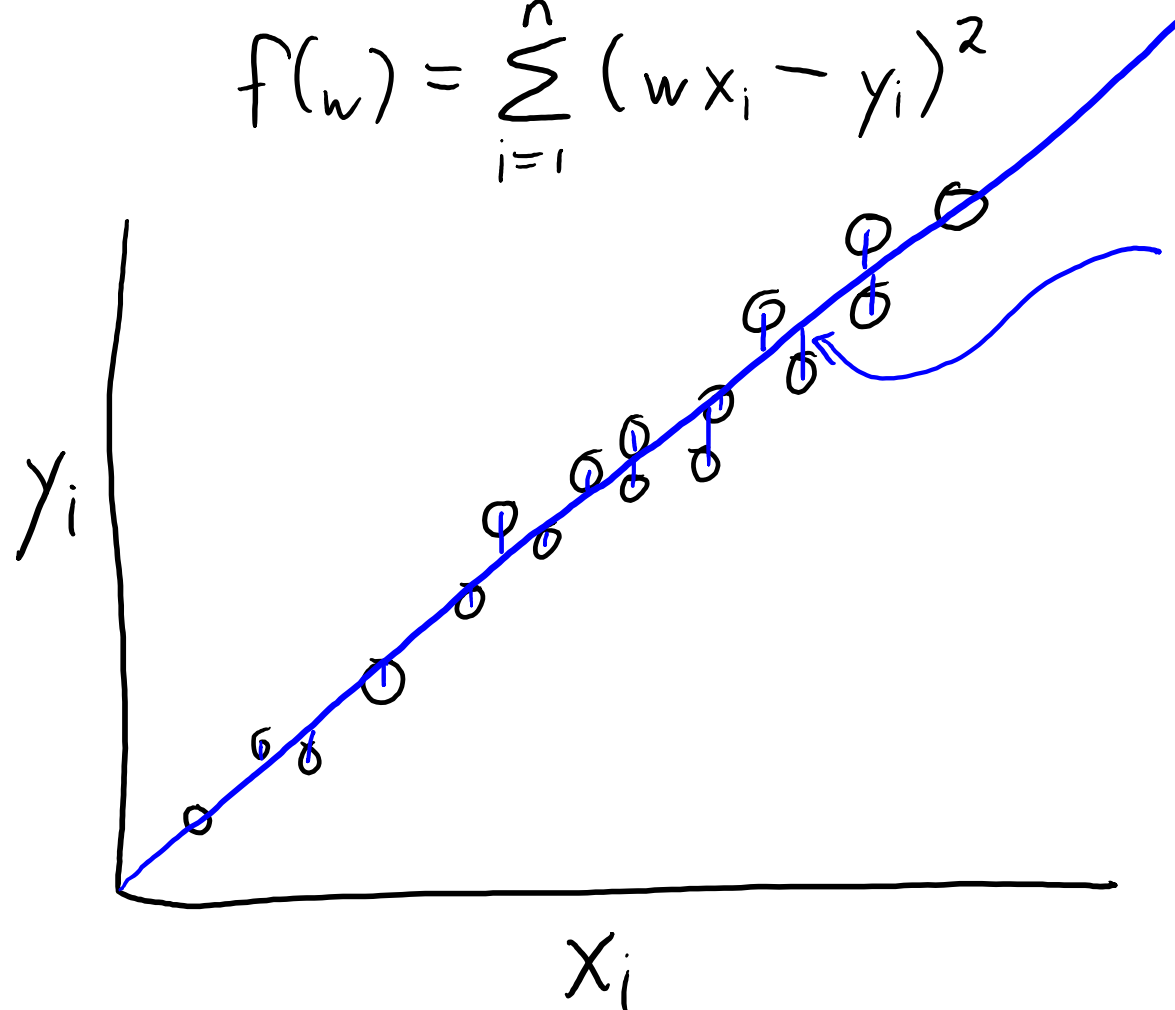
(residual)
Difference between prediction and true value for example 'i'.

- If this is zero, we exactly fit data. If this small, line is "close" to data.
- There are some justifications for choosing this function 'f'.
 - A probabilistic interpretation is coming later in the course.
- But usually, we choose this 'f' because **it is easy to minimize**.

Linear Least Squares Objective Function

- Linear least squares sets 'w' is to minimize **sum of squared residuals**:

$$f(w) = \sum_{i=1}^n (wx_i - y_i)^2$$



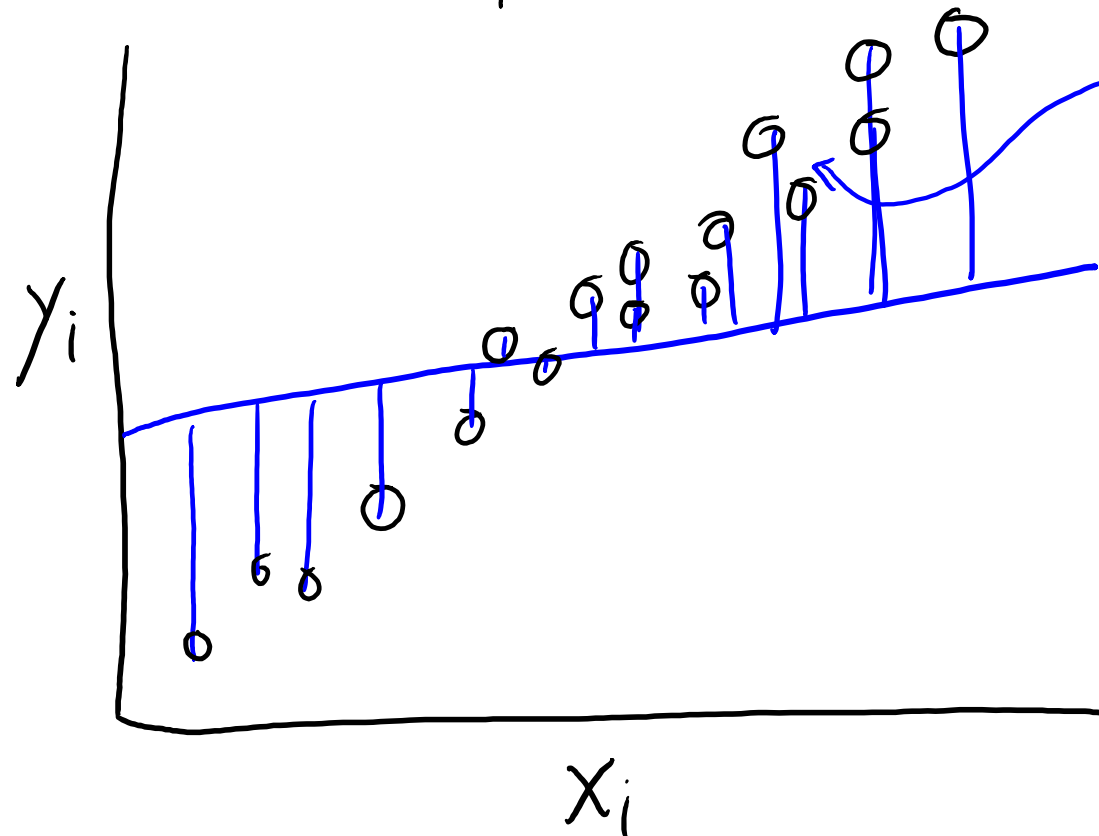
"Error" is the sum of the squared values of these vertical distances between the line ($w x_i$) and the targets (y_i)

↓
If this error is small, then our predictions are close to the targets.

Linear Least Squares Objective Function

- Linear least squares sets 'w' is to minimize **sum of squared residuals**:

$$f(w) = \sum_{i=1}^n (wx_i - y_i)^2$$

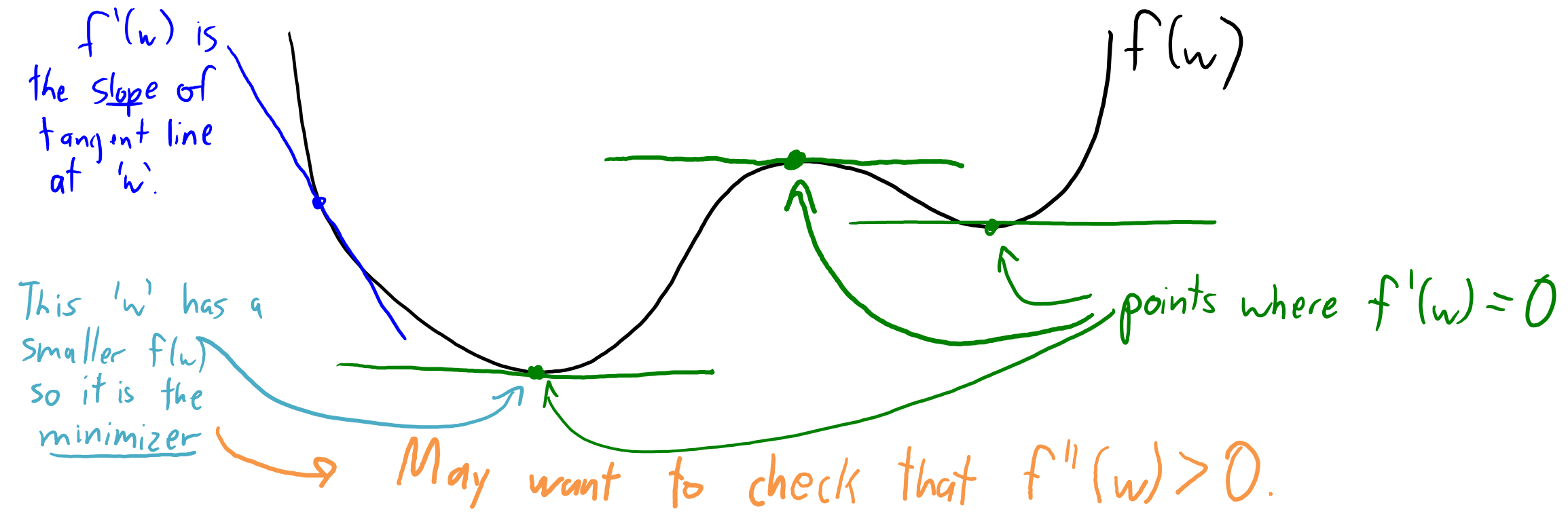


"Error" is the sum of the squared values of these vertical distances between the line ($w x_i$) and the targets (y_i)

↓
If this error is **large**, then our predictions are **far from** the targets.

Minimizing a Differentiable Function

- Math 101 approach to minimizing a differentiable function 'f':
 - Take the derivative of 'f'.
 - Find points 'w' where the derivative $f'(w)$ is equal to 0.
 - Choose the smallest one (and check that $f''(w)$ is positive).



Digression: Multiplying by a Positive Constant

- Note that this problem:

$$f(w) = \sum_{i=1}^n (w x_i - y_i)^2$$

- Has the **same set of minimizers** as this problem:

$$f(w) = \frac{1}{2} \sum_{i=1}^n (w x_i - y_i)^2$$

- And these also have the same minimizers:

$$f(w) = \frac{1}{n} \sum_{i=1}^n (w x_i - y_i)^2 \quad f(w) = \frac{1}{2n} \sum_{i=1}^n (w x_i - y_i)^2 + 1000$$

- I can **multiply 'f' by any positive constant and not change solution.**
 - Derivative will still be zero at the same locations.
 - We will use this trick a lot!

Deriving Least Squares Solution

$$\begin{aligned}
 f(w) &= \frac{1}{2} \sum_{i=1}^n (w x_i - y_i)^2 = \frac{1}{2} \sum_{i=1}^n [w^2 x_i^2 - 2w x_i y_i + y_i^2] && \text{(expand square)} \\
 &= \frac{1}{2} \sum_{i=1}^n w^2 x_i^2 - \frac{1}{2} \sum_{i=1}^n 2w x_i y_i + \frac{1}{2} \sum_{i=1}^n y_i^2 && \text{(split sum up into different terms)} \\
 &= \underbrace{\frac{w^2}{2} \sum_{i=1}^n x_i^2}_{\text{constant 'a'}} - w \underbrace{\sum_{i=1}^n x_i y_i}_{\text{constant 'b'}} + \underbrace{\frac{1}{2} \sum_{i=1}^n y_i^2}_{\text{constant 'c'}} && \text{(remove factors from sums that do not depend on example 'i')} \\
 &= \frac{w^2}{2} a - wb + c
 \end{aligned}$$

Take derivative: $f'(w) = wa - b + 0$

Setting $f'(w) = 0$ and solving gives $w = \frac{b}{a} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$ (exists if we have a non-zero feature)

Finding Least Squares Solution

- Finding 'w' that minimizes **sum of squared errors**:

Setting $f'(w)=0$ and solving gives $w = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$ (exists if we have one non-zero x_i)

- Let's check that this is a **minimizer** by checking second derivative:

$$f'(w) = w \sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i y_i$$

$$f''(w) = \sum_{i=1}^n x_i^2$$

- Since (anything)² is non-negative, we have $f''(w) \geq 0$.
- If at least one feature is not zero, then $f''(w) > 0$ and 'w' is a minimizer.

Next Topic: Least Squares in 'd' Dimensions

Motivation: Combining Explanatory Variables

- Smoking is **not the only contributor** to lung cancer.
 - For example, there environmental factors like exposure to asbestos.
- How can we model the **combined effect** of smoking and asbestos?
- A simple way is with a **2-dimensional linear function**:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$$

"weight" of feature 1 (points to w_1)
 Value of feature 1 in example 'i' (points to x_{i1})
 "weight" on feature 2 (points to w_2)
 Value of feature 2 in example 'i' (points to x_{i2})

- We have a weight w_1 for feature '1' and w_2 for feature '2':

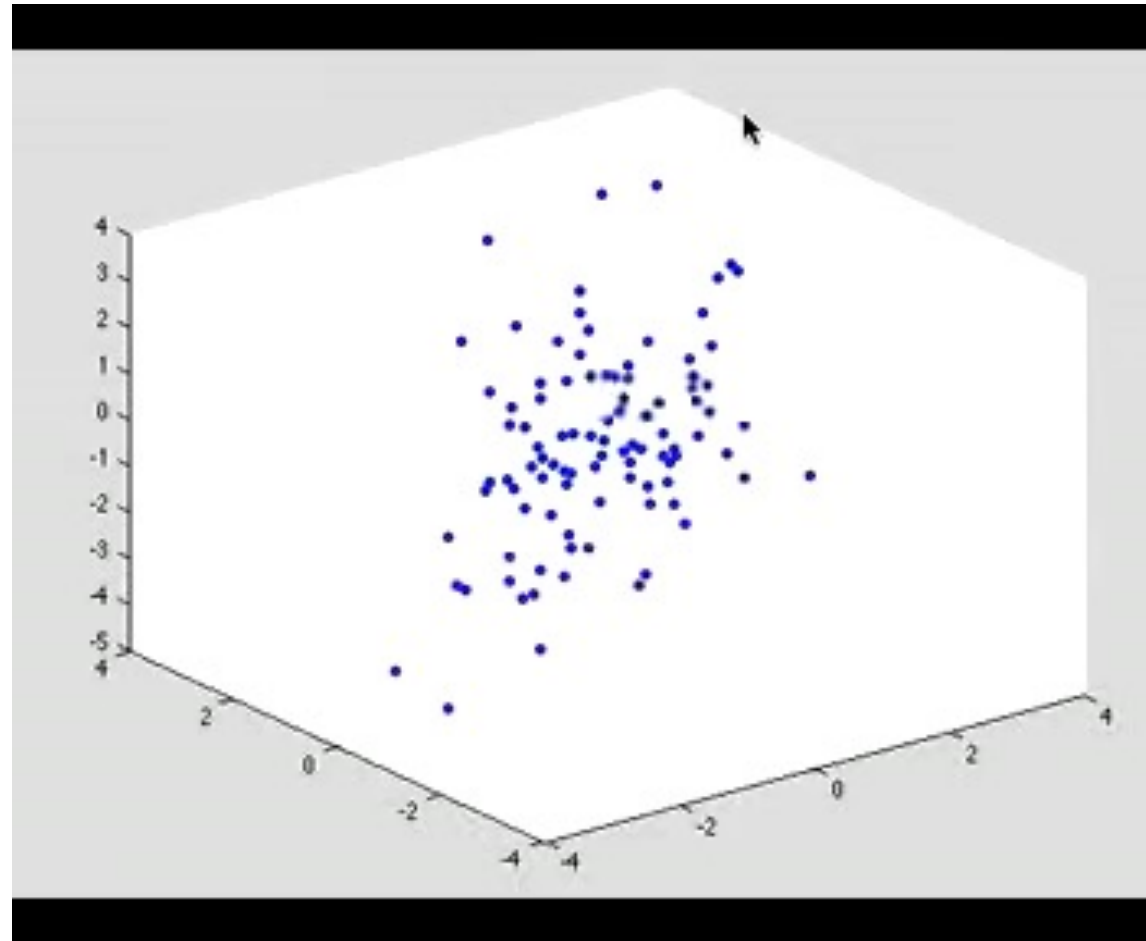
$$\hat{y}_i = 10(\# \text{ cigarettes}) + 25(\# \text{ asbestos})$$

Linear Regression in 2-Dimensions

- Linear model:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$$

- This defines a **two-dimensional plane**.

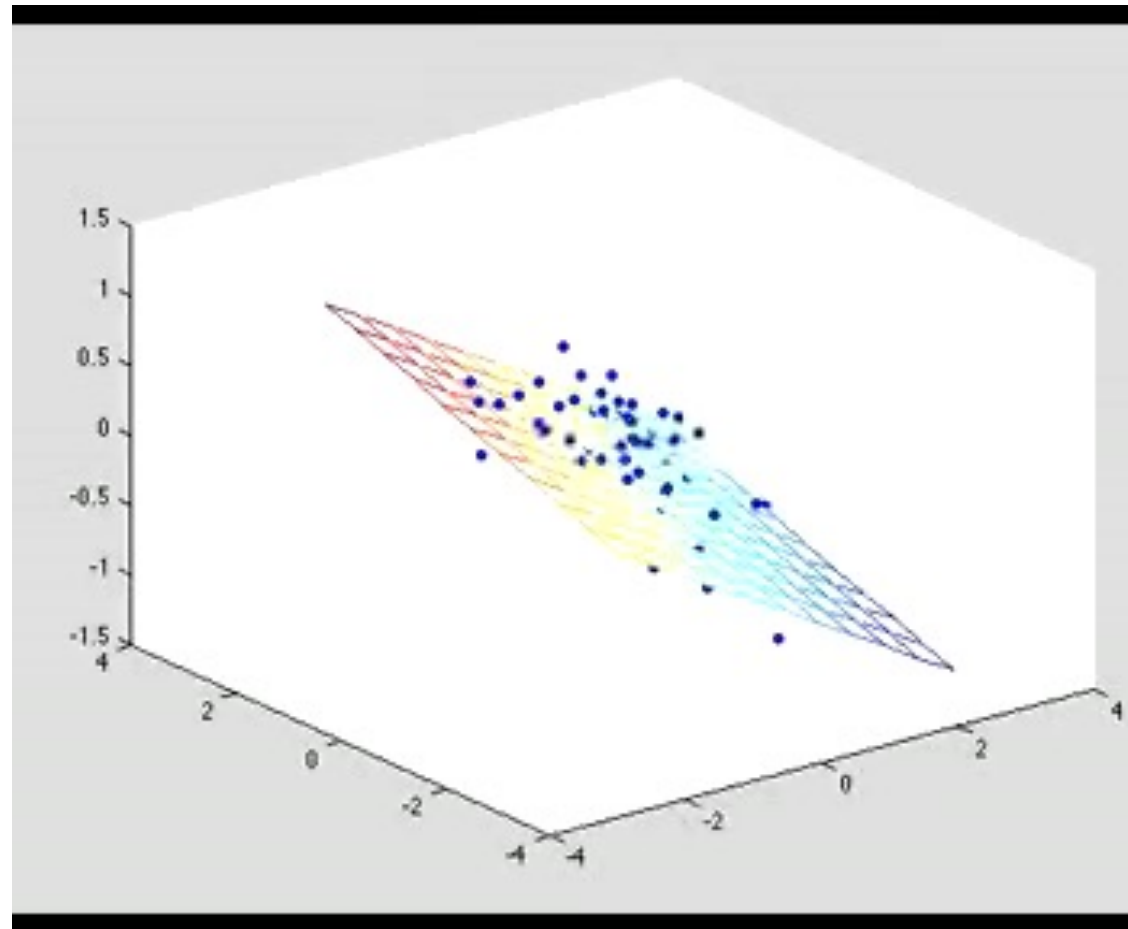


Linear Regression in 2-Dimensions

- Linear model:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$$

- This defines a
**two-dimensional
plane.**
- **Not just a line!**



Linear Regression in d-Dimensions

- If we have 'd' features, the **d-dimensional linear model** is:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + \dots + w_d x_{id}$$

- In words, **prediction is a weighted sum of the features**.
- We can re-write this using summation notation as:

$$\hat{y}_i = \sum_{j=1}^d w_j x_{ij}$$

- We can again choose 'w' to **minimize the sum of squared** residuals:

$$f(w_1, w_2, \dots, w_d) = \frac{1}{2} \sum_{i=1}^n \left(\underbrace{\sum_{j=1}^d w_j x_{ij}}_{\hat{y}_i} - y_i \right)^2$$

- Dates back to 1801: Gauss used it to predict location of the asteroid Ceres.
 - We can use multi-variable calculus to minimize 'f' with respect to the parameters $w_1, w_2, \dots, w_d$.

Minimizing Multi-Variable Differentiable Function

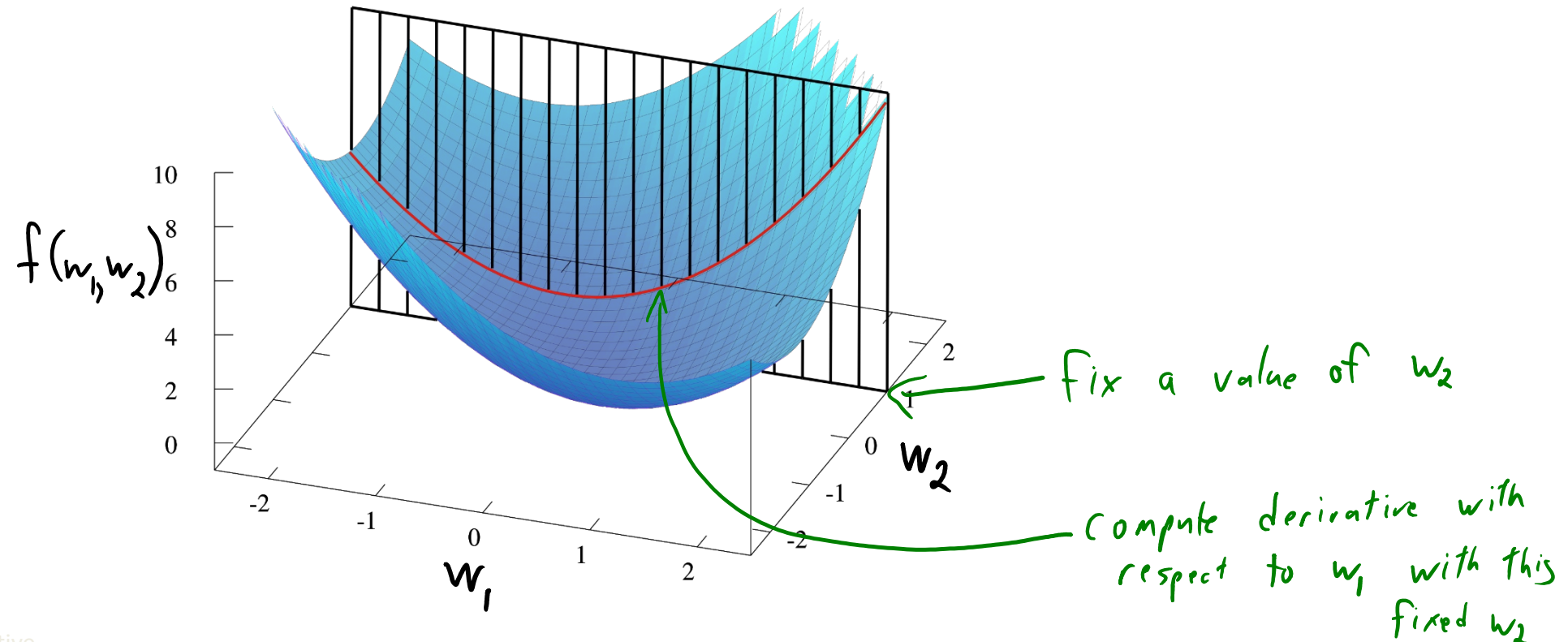
- With one variable, we “find ‘w’ where the derivative is equal to 0”.
- The generalization of this idea to when we have ‘d’ variables:
 - “Find ‘w’ where the **gradient vector** is **equal to the zero vector**”.
- **Gradient** is a **vector with partial derivative ‘j’** in position ‘j’.

$$\underbrace{\nabla f(w)}_{\text{gradient vector}} = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \vdots \\ \frac{\partial f}{\partial w_d} \end{bmatrix}$$

} partial derivative of 'f' with respect to variable w_2

Review: Partial Derivative

- Partial derivative with respect to w_j (written $\frac{\partial f}{\partial w_j}$).
 - Derivative with respect to w_j , keeping all others variables fixed.



Partial Derivative for Least Squares

- Partial derivative with respect to w_1 for **least squares** with $n=1$:

$$f(w_1, w_2, \dots, w_d) = \frac{1}{2} \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2$$

$$\frac{\partial f}{\partial w_1} = \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i1}$$

$$\sum_{j=1}^d w_j x_{ij} = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

only one term involving w_1

$$\frac{\partial}{\partial w_1} \left[\frac{1}{2} \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2 \right] = \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) \frac{\partial}{\partial w_1} \left[\sum_{j=1}^d w_j x_{ij} - y_i \right]$$

$$\frac{\partial}{\partial w_1} \left[\frac{1}{2} g(w_1, w_2, \dots, w_d)^2 \right] = g(w_1, w_2, \dots, w_d) \frac{\partial}{\partial w_1} [g(w_1, w_2, \dots, w_d)]$$

$$= \frac{\partial}{\partial w_1} [w_1 x_{i1} + \text{terms with no } w_1] = x_{i1} + 0$$

Partial Derivative for Least Squares

- Partial derivative with respect to w_j for **least squares** with $n=1$:

$$f(w_1, w_2, \dots, w_d) = \frac{1}{2} \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2$$

$$\frac{\partial f}{\partial w_j} = \left(\sum_{j'=1}^d w_{j'} x_{ij'} - y_i \right) x_{ij}$$

→ We use j' to distinguish the summation index from the variable w_j we are differentiating

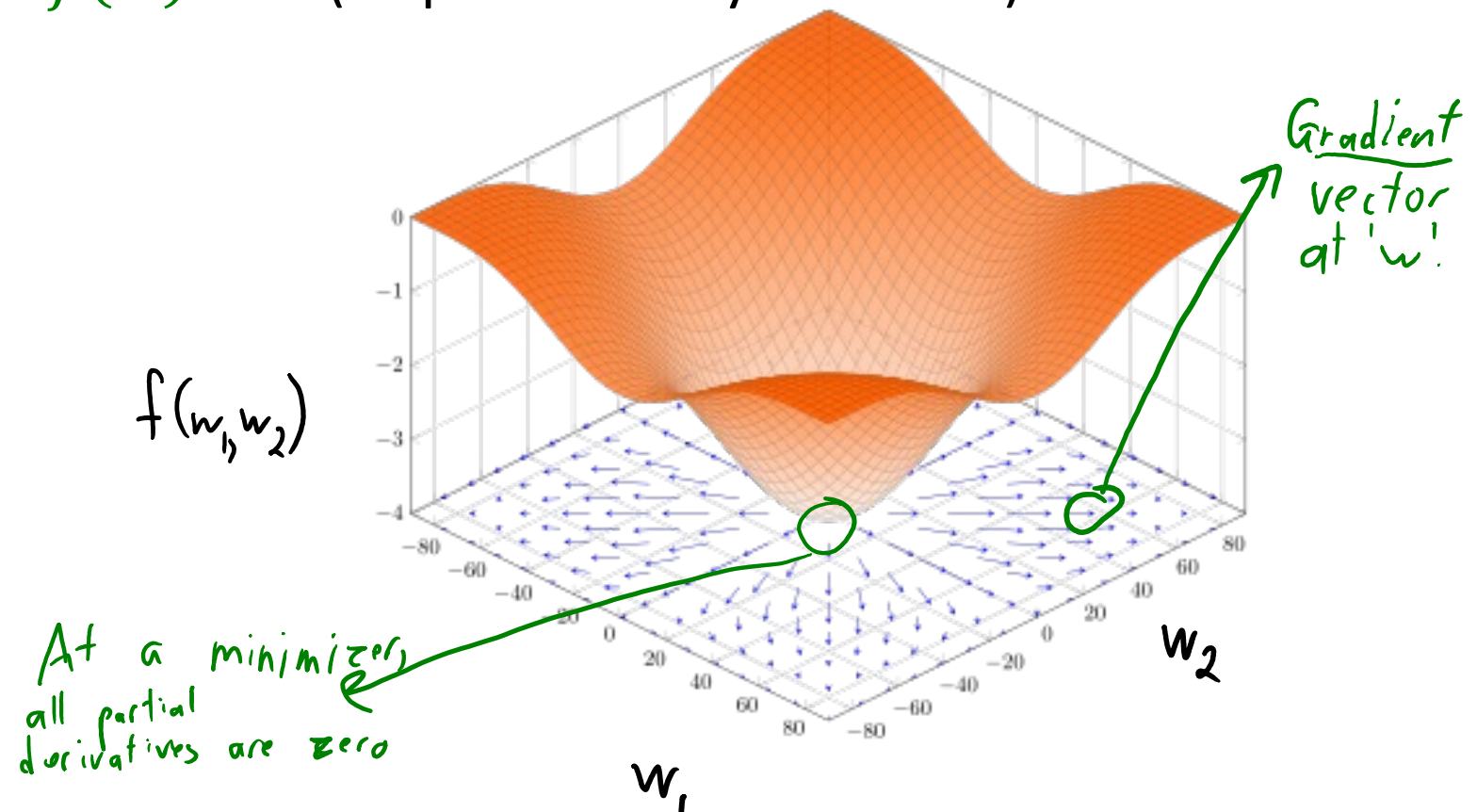
- Partial derivative with respect to w_j for **least squares** for general 'n':

$$f(w_1, w_2, \dots, w_d) = \frac{1}{2} \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2$$

$$\frac{\partial f}{\partial w_j} = \sum_{i=1}^n \left(\sum_{j'=1}^d w_{j'} x_{ij'} - y_i \right) x_{ij}$$

Review: Gradient Vector

- The **gradient vector** is the concatenation of all partial derivatives:
 - At 'w', $\nabla f(w)$ is in the direction with most-positive slope.
 - At minimizers we have $\nabla f(w) = 0$ (slope is 0 every direction).



Gradient Vector for Least Squares

- By concatenating partial derivatives, for least squares gradient is:

$$\nabla f(\mathbf{w}) = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \vdots \\ \frac{\partial f}{\partial w_d} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i1} \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i2} \\ \vdots \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{id} \end{bmatrix}$$

- So to train a least squares model, we need this to **equal the zero vector**.

Fitting a Linear Least Squares Model

- Setting **gradient to equal 0 vector** for linear least squares gives:

$$\nabla f(\mathbf{w}) = 0 \iff \begin{aligned} \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i1} &= 0 \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i2} &= 0 \\ &\vdots \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{id} &= 0 \end{aligned}$$

- This is a set of 'd' **linear equations**, with 'd' unknowns ($w_1, w_2, \dots, w_d$).
 - You can solve these equations using **Gaussian elimination** (linear algebra).
- Claim: **all 'w' with $\nabla f(\mathbf{w}) = 0$ are minimizers** (we will discuss why later).
 - May be more than one 'w' satisfying this, but all have the same minimum error.

Next Topic: Matrix Notation

Matrix Notation: Motivation

- We have expressed linear least squares with **summation notation**:

$$f(w_1, w_2, \dots, w_d) = \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2$$

- But you often see it equivalently expressed using **matrix notation**:

$$f(w) = \|Xw - y\|^2$$

- Why do people use matrix notation?
 - Can be easier to understand and lead to “nicer” code (once you are used to it).
 - Makes it easier to see some properties (like the connection to norms above).
 - Or derive properties, like showing that all ‘w’ with $\nabla f(w) = 0$ are minimizers.
 - Can lead to **code with fewer bugs**.
 - Since you can use existing implementations of standard operations.
 - Can lead to **faster code**.
 - If we are using packages that implement fast matrix operations.

Matrix Notation (MEMORIZE/STUDY THIS)

- Instead of lists, **we are now going to treat features x_i as vectors.**
 - And we need to decide if it will be a 1xd or a dx1 vector.
- In this course, all **vectors are assumed to be column-vectors:**

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix}$$

- So **rows of matrix 'X' will be transposes of the column-vectors x_i :**

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nd} \end{bmatrix} = \begin{bmatrix} \text{---} x_1^T \text{---} \\ \text{---} x_2^T \text{---} \\ \vdots \\ \text{---} x_n^T \text{---} \end{bmatrix}$$

Matrix Notation (MEMORIZE/STUDY THIS)

- Linear regression **prediction for one example** in matrix notation:

$$\hat{y}_i = w^T x_i$$

- Why?

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id} = \underbrace{[w_1 \ w_2 \ \dots \ w_d]}_{w^T} \underbrace{\begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix}}_{x_i} = w^T x_i$$

- Using $\hat{y}_i = w^T x_i$, we can re-write sum of squared residuals as:

$$\sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (w^T x_i - y_i)^2$$

Matrix Notation (MEMORIZE/STUDY THIS)

- Linear regression **prediction for all 'n' example** in matrix notation:

$$\hat{y} = Xw$$

$n \times 1$ vector of predictions $n \times d$ matrix of features $d \times 1$ vector of parameters.

- Why?

$$Xw = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix} \begin{bmatrix} 1 \\ w \\ 1 \end{bmatrix} = \begin{bmatrix} x_1^T w \\ x_2^T w \\ \vdots \\ x_n^T w \end{bmatrix} = \begin{bmatrix} w^T x_1 \\ w^T x_2 \\ \vdots \\ w^T x_n \end{bmatrix} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \hat{y}$$

We have $w^T x_i = x_i^T w$ because it is a scalar.
 (For example, $[5]^T = [5]$)

Prediction for example 'i' in row 'i'

Matrix Notation (MEMORIZE/STUDY THIS)

- Linear regression **residual vector** in matrix notation:

$$r = X_w - y$$

$\rightarrow n \times 1$ vector of residuals $r_i = \hat{y}_i - y_i$

- Why?

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix} = \begin{bmatrix} \hat{y}_1 - y_1 \\ \hat{y}_2 - y_2 \\ \vdots \\ \hat{y}_n - y_n \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}}_{\hat{y}} - \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_y = \hat{y} - y = X_w - y$$

$\hat{y} = X_w$ (last slide)

Matrix Notation (MEMORIZE/STUDY THIS)

- Different ways to write **sum of residuals squared** in linear regression model:

$$\begin{aligned}
 f(w) &= \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right)^2 \\
 &= \sum_{i=1}^n (w^T x_i - y_i)^2 \\
 &= \sum_{i=1}^n r_i^2 \\
 &= \|r\|^2 \\
 &= \|Xw - y\|^2
 \end{aligned}$$

Can also write $\|r\|^2 = r^T r$
 Can also write $\|Xw - y\|^2$
 $= (Xw - y)^T (Xw - y)$
 or $\|y^{\wedge} - y\|^2$

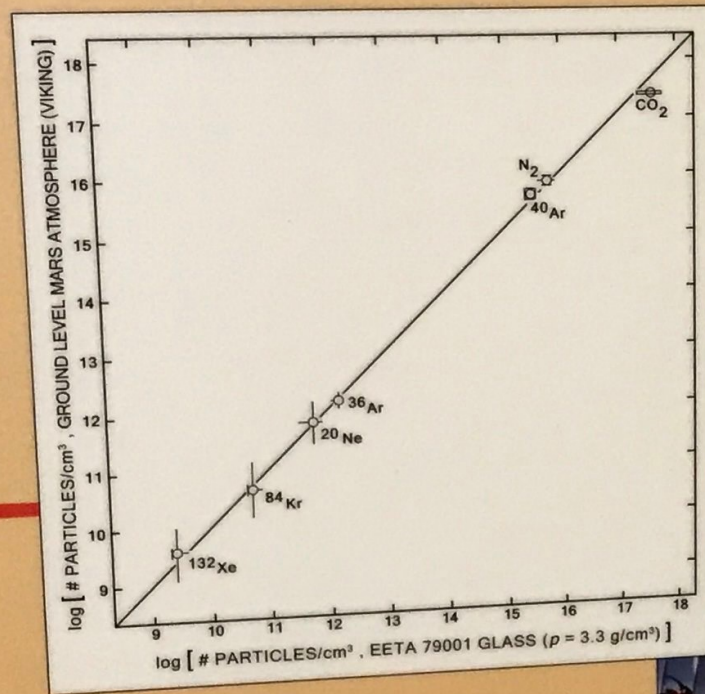
- So least squares **minimizes L2-norm** between target and predictions.

Summary

- **Regression** considers the case of a numerical y_i .
- **Least squares** is a classic method for fitting linear models.
 - Minimizes sum of squared residuals (prediction and true value difference).
 - With 1 feature, it has a simple closed-form solution.
 - Can be generalized to 'd' features, taking linear weighting of features.
- **Gradient** is vector containing partial derivatives of all variables.
- **Matrix notation** for expressing least squares problem: $\|Xw - y\|^2$.
- Next time:

minimizing $\frac{1}{2} \sum_{i=1}^n (w^T x_i - y_i)^2$ in terms of 'w' using 1 line of code

- In Smithsonian National Air and Space Museum (Washington, DC):



Scientists found in the meteorite trapped gas whose composition was nearly identical to the Martian atmosphere as measured by the Viking Landers. This graph compares the concentration of gases in the Martian atmosphere (vertical axis) with their concentration in the meteorite (horizontal axis). If they matched perfectly, the points would fall on the diagonal line. The close match strongly suggests that this meteorite came from Mars.



Causality, Interventions, and RCTs

- What if you want to assess **causality**?
- You can sometimes do this by **collecting data in specific ways**.
 - You need to set the values of the features “by **intervention**”.
 - You do not passively observe, you **set** them and then watch the effect.
 - Most common way this is done is with a **randomized control trial**.
 - Say you want to evaluate the effectiveness of a pill for a certain disease.
 - You get a bunch of people with the disease for training data.
 - You randomly decide which of the people will take the pill, and which won't.
 - If the people who got the pill did better/worse on average, it was caused by the pill.
 - The randomness takes away the possibility that certain groups are more/less likely to take the pill.
 - Group not taking the pill often given placebo, removing effect of “feel like you are being treated”.
 - Often the researchers do not even get to know who took the pills until after the study is over.
 - » “Double blind”, to avoid the researchers giving hints about who got the pill.

Converting Partial Derivative to Matrix Notation

- Re-writing linear least squares **partial derivative in matrix notation**:

$$\begin{aligned}
 \frac{\partial f}{\partial w_j} &= \sum_{i=1}^n \left(\sum_{j'=1}^d w_{j'} x_{ij'} - y_i \right) x_{ij} && \text{(from earlier)} \\
 &= \sum_{i=1}^n (w^T x_i - y_i) x_{ij} && \text{(no need for } j') \\
 &= \sum_{i=1}^n r_i x_{ij} && \text{(definition of } r_i) \\
 &= r^T x^j
 \end{aligned}$$

residual vector $\swarrow$ column ' j ' of ' X '

Converting Gradient to Matrix Notation

- Re-writing linear least squares **gradient in matrix notation**:

$$\nabla f(w) = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \vdots \\ \frac{\partial f}{\partial w_d} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i1} \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{i2} \\ \vdots \\ \sum_{i=1}^n \left(\sum_{j=1}^d w_j x_{ij} - y_i \right) x_{id} \end{bmatrix} = \begin{bmatrix} r^T x^1 \\ r^T x^2 \\ \vdots \\ r^T x^d \end{bmatrix} \quad (\text{from last slide})$$

$$= \underbrace{\begin{bmatrix} \text{---} x^1 \text{---} \\ \text{---} x^2 \text{---} \\ \vdots \\ \text{---} x^d \text{---} \end{bmatrix}}_{X^T} r = X^T r$$

Transpose of 'X' multiplied by residual vector